Aligned Mathematical Structures

Source-backed grouped formula sample

1 Expressions

Expression 1. The following expression is taken from a source-backed formula corpus.

$$\begin{aligned} & \Delta^{(1)}_{(n-1l+1)}(x_1,\cdots,x_m|z_1,\cdots,z_{n-1}|z_n,\cdots,z_N) \\ = \quad & q^{l-n+1}\sum_{k=n}^{N} z_k \frac{\prod_{j=1}^{n-1}(z_k-z_jq^{-2})}{\prod_{\substack{i=n \\ i\neq k}}^{N}(z_i-z_k)q^{-1}} \Delta^{(nl)}(x_1,\cdots,x_m|z_1,\cdots,z_{n-1},z_k|z_n,\overset{k}{\hat{\cdots}},z_N), \end{aligned}$$

Expression 2. The following expression is taken from a source-backed formula corpus.

$$\begin{aligned} & [\alpha_{\mathcal{L}}(a),\alpha_{\mathcal{L}}(b),[x,y,z]_{\mathcal{L}}]_{\mathcal{L}} \\ & = [[a,b,x]_{\mathcal{L}},\alpha_{\mathcal{L}}(y),\alpha_{\mathcal{L}}(z)]_{\mathcal{L}} + [\alpha_{\mathcal{L}}(x),[a,b,y]_{\mathcal{L}},\alpha_{\mathcal{L}}(z)]_{\mathcal{L}} + [\alpha_{\mathcal{L}}(x),\alpha_{\mathcal{L}}(y),[a,b,z]_{\mathcal{L}}]_{\mathcal{L}}, \end{aligned}$$

Expression 3. The following expression is taken from a source-backed formula corpus.

$$(II) \quad \begin{pmatrix} x_1+\mathrm{i}y_1 \\ x_2+\mathrm{i}y_2 \\ x_3+\mathrm{i}y_3 \\ x_4+\mathrm{i}y_4 \end{pmatrix} \quad \leftrightarrow \quad \frac{1}{\sqrt{2}} \begin{pmatrix} -(y_2+y_4)+\mathrm{j}(y_2-y_4) \\ (y_1+y_3)-\mathrm{j}(y_1-y_3) \\ (x_2+x_4)-\mathrm{j}(x_2-x_4) \\ -(x_1+x_3)+\mathrm{j}(x_1-x_3) \end{pmatrix},$$

Expression 4. The following expression is taken from a source-backed formula corpus.

$$\begin{aligned} & (|\partial_t^\ell \mathcal{D}^\alpha \phi|^2)_t \\ & beginalign*8pt] = -2\sum_{\ell_1+|\alpha_1|>0} \partial_t^\ell \mathcal{D}^\alpha\phi \partial_t^{\ell_1}\mathcal{D}^{\alpha_1} v\cdot\nabla(\partial_t^{\ell_2}\mathcal{D}^{\alpha_2}\phi) - v\cdot\nabla|\partial_t^\ell\mathcal{D}^\alpha\phi|^2. \\ & beginalign*6pt] \end{aligned}$$

Expression 5. The following expression is taken from a source-backed formula corpus.

$$\begin{aligned} \mathrm{Tr}\left[\Gamma_s(P^a)\Gamma_s(P_b)\right] &= \delta^a{}_b\,, \qquad \Rightarrow \quad \Gamma_s(P^a) = \tfrac{-i}{2g}\gamma^a\,, \\ \mathrm{Tr}\left[\Gamma_s(M^{ab})\Gamma_s(P_{cd})\right] &= \delta^{ab}{}_{cd}\,, \qquad \Rightarrow \quad \Gamma_s(M^{ab}) = -\tfrac{1}{2}\gamma^{ab}\,. \end{aligned}$$